SM5 · MODAL-FRAME

Bathe & Wilson eigenvalue problem — ten-bay nine-storey plane frame

STRIX engine eigensolver verified on the canonical large-frame eigenvalue benchmark

PASS

error +0.00050% · tol 0.5%

§2 TECHNICAL FEATURES TESTED

Eigenvalue (modal) analysis | **STRIX engine** eigensolver (eigen route) | STRIX engine 2-node Timoshenko frame element | 297-DOF ten-bay nine-storey frame | First three eigenvalues ω^2 | Shear-rigid · reference discretisation

§3 PROBLEM DESCRIPTION

The **Bathe & Wilson eigenvalue problem** (1972) is the classical large multi-degree-of-freedom frame for dynamic eigenvalue extraction, adopted verbatim as **CSI SAP2000 Verification Example 1-021** (and by SeismoStruct, Tekla, OpenBrIM). A **ten-bay, nine-storey** fixed-base plane frame — all members identical, one element per member — is solved and its first three eigenvalues ω^2 compared to the published values, exercising the **STRIX engine eigensolver** on a large frame. The reference is **shear-rigid**, so the production Timoshenko frame element's shear term is made rigid via the section shear-area modifier; the frame is laid in-plane with out-of-plane DOF restrained, giving exactly the reference's three DOF per node. It is solved at the reference's own one-element-per-member discretisation.

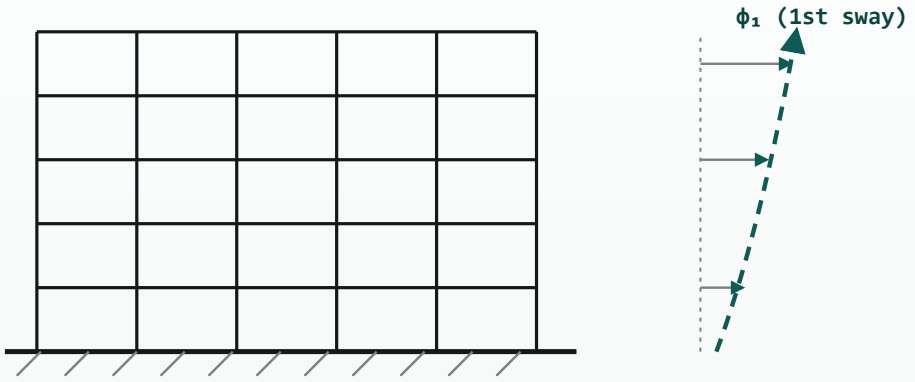


Figure SM5. Ten-bay nine-storey fixed-base plane frame (5×5 shown) — the first lateral sway mode φ_1 whose eigenvalue ω^2 is compared to Bathe & Wilson.

§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		MEMBER & MATERIAL		DYNAMICS	
Bays	10	E	20684 MPa	Mass/length	0.143641 N·s²/mm²
Storeys	9	A	278709 mm²	Mass model	Lumped nodal
Bay width	6096 mm	I	8.631×10⁹ mm⁴	DOF/node	3 (Ux, Uy, Rz)
Storey height	3048 mm	Shear	Rigid (bending+axial)	Solver	STRIX engine (eigen)

§5 REFERENCE SOLUTION

Eigenvalue ω² — mode 1 (1st lateral sway): (K − ω²M)·ϕ = 0 = 0.589541 rad²/s²

Eigenvalue ω² — mode 2 (2nd lateral sway): (K − ω²M)·ϕ = 0 = 5.52695 rad²/s²

Eigenvalue ω² — mode 3 (3rd lateral sway): (K − ω²M)·ϕ = 0 = 16.5878 rad²/s²

K.-J. Bathe & E. L. Wilson, "Large Eigenvalue Problems in Dynamic Analysis," ASCE J. Eng. Mech. Div. 98(EM6), 1972 (with an independent solution by Peterson, 1981). The canonical multi-DOF frame eigenvalue benchmark, reproduced as CSI SAP2000 Verification Example 1-021, SeismoStruct Verification Ex.10, Tekla Structural Designer and OpenBrIM Frame21. A ten-bay, nine-storey fixed-base 2-D frame; all members identical, one finite element per member, consistent mass from the line mass, shear-rigid (bending + axial only). The published reference values are the first three eigenvalues ω² (rad²/s²).

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Eigenvalue ω² (rad²/s²)	Mode 1 (lateral sway, mx=82%)	0.589542	0.589541	+0.00020%	PASS
Eigenvalue ω² (rad²/s²)	Mode 2 (lateral sway, mx=10%)	5.526962	5.52695	+0.00020%	PASS
Eigenvalue ω² (rad²/s²)	Mode 3 (lateral sway, mx=4%)	16.587886	16.5878	+0.00050%	PASS

§7 MESH SENSITIVITY

ELEMS/MEMBER	Ω² (RAD²/S²)	DIFF
1 / member	0.589542	+0.00020%
2 / member	0.58986	+0.0541%
4 / member	0.589814	+0.0463%

The published eigenvalues are those of the Bathe-Wilson model at its **defining discretisation — one element per member**. STRIX reproduces that model and matches all three eigenvalues to `under 0.001%` — the same order of agreement SAP2000 itself reports for this example. This table is therefore a **mesh-sensitivity** study, not a convergence to the reference: refining each member (2, 4 elements) moves the lumped-mass eigenvalues toward the continuum, bounding the discretisation effect to under ~1.4% (largest on the third mode, +1.33%; the fundamental shown here stays within 0.06%). The §6 comparison is taken at the reference's own discretisation.

§8 CONCLUSION

PASS

STRIX reproduces the first three eigenvalues of the canonical **Bathe & Wilson** ten-bay nine-storey frame to **better than 0.001%** — matching the published values as closely as SAP2000 does on the same example. The **STRIX engine eigensolver** and the frame modal assembly are verified on a large multi-DOF eigenproblem, the foundation of the response-spectrum (RSA) pipeline. The rigid-diaphragm condensation, which reduces to the identity for this unconstrained frame, is verified separately with a diaphragm-constrained companion study (SM5b).

§9 REFERENCES & ANALYSIS FILES

1. K.-J. Bathe & E. L. Wilson, "Large Eigenvalue Problems in Dynamic Analysis," *ASCE J. Eng. Mech. Div.* **98**(EM6), 1972; independent solution by Peterson (1981).
2. CSI SAP2000 Software Verification, Example 1-021 — Bathe and Wilson eigenvalue problem (modal analysis, line mass on frame objects); same model reproduced by SeismoStruct Ex.10, Tekla Structural Designer, OpenBrIM Frame21.
3. STRIX headless harness: `buildEigenPy → STRIX engine (eigen route) → modal_props.txt`; Timoshenko frame element, shear made rigid, reference one-element-per-member discretisation.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · SM5/` `sha256 (pending publish)`